

# **Sample Document for GenAI PDF Reader Project**

## **Introduction to Artificial Intelligence**

Artificial Intelligence (AI) refers to the simulation of human intelligence in machines. AI systems are designed to perform tasks such as reasoning, learning, problem solving, perception, and language understanding. Examples include recommendation systems, chatbots, and self-driving cars.

## **Machine Learning**

Machine Learning (ML) is a subset of AI that allows systems to learn patterns from data without being explicitly programmed. Common types include supervised learning, unsupervised learning, and reinforcement learning.

## **Large Language Models**

Large Language Models (LLMs) such as GPT models are trained on massive datasets to understand and generate human-like text. They are widely used in chatbots, document summarization, question answering, and code generation.

## **What is RAG**

Retrieval Augmented Generation (RAG) is a technique where an LLM retrieves relevant documents from a knowledge base and uses them as context to generate accurate answers. This approach helps reduce hallucinations and improves factual accuracy.

## **LangChain Overview**

LangChain is a framework used to build applications powered by large language models. It provides tools for prompt management, chains, agents, memory, and document loaders. Developers often use LangChain for building chatbots, document Q&A systems, and AI agents.

## **FastAPI**

FastAPI is a modern Python web framework used for building APIs quickly and efficiently. It supports asynchronous programming and automatic API documentation using Swagger UI. FastAPI is commonly used to expose machine learning or AI models as APIs.

## Key Concepts Summary

| Concept          | Description                                           |
|------------------|-------------------------------------------------------|
| AI               | Simulation of human intelligence in machines          |
| Machine Learning | Algorithms that learn patterns from data              |
| LLM              | Large Language Model trained on huge datasets         |
| RAG              | Combines retrieval with generation for better answers |